Coding & Solution

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Name	Rising Waters (Flood Prediction System)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Bhavana-229/Rising-Waters
Programming Language(s)	Python,Javascript
Framework(s) Used	Streamlit, Scikit-learn, Pandas, NumPy
Key Features Implemented	Flood Prediction, Data Processing, Visualization, Risk Classification, Alert Generation
Pending / Incomplete Features	Real-time IoT Sensor Integration
Setup / Run Instructions	Install requirements and run `streamlit run app.py`

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The solution follows modular programming principles and uses
machine learning techniques for flood prediction with clear documentation.